

CSE260 Assignment 03
[C03]
Total Mark: 30
Deadline: August 23, 2025; 11:59PM
Submission Link: [Click Here](#)

This assignment must be hand-written. Show ALL steps in ALL questions.

1. Design a 10 person voting system using Half Adder and 4bit parallel adder. Your circuit must be efficient. [5 Mark]
2. Implement the following boolean function using (a) 16:1 mux (c) 8:1 mux d) 4:16 decoder e) 3:8 decoders f) 2:4 decoders
$$F(A,B,C,D) = \Sigma(0,1,2,7,8,10,11,13, 15)$$

Make sure that your circuit is efficient, meaning you should use the lowest number of components. [10 Mark]
3. Design a BCD to Excess 5 code converter using 4:16 decoder and 16:4 encoder. [5 Mark]
4. Implement full adder using encoder and decoder. [5 Mark]
5. Design a 19 person voting system using encoder(s) and decoder(s) and parallel adder(s). You can not use full adders. Meaning, for the inputs, you MUST use encoder(s) and decoder(s) as required. Make sure your circuit is optimized. [5 Mark]